Question 1

a)

i) aliasing means higher frequency components of the signal cannot be captured because of the low sampling frequency, and leads to overlap in the spectrum. The result of aliasing is that the produced sound will be in poor quality and the original data cannot be produced.

Avoid aliasing: know the maximum frequency of the sound wave, then set the sampling rate as 2 times higher than the maximum frequency of the sound wave.

ii) 2f. f is the max f of the sound wave.

iii)

image: 200 * 300 * 24 * 24 * 60 / 8 = 259200000 bytes

audio: 15000 * 8 * 60 / 8 = 900000 bytes

video = image + audio = 260100000 bytes

b)

i) grayscale image uses 8-bit integers to store each pixel value, which means the grayscale levels range from 0 to 255. However, the true color image forms a 24-bit pixel palette which has 16.7 million colors. Therefore, a grayscale image contains 3 times less data than a true color image.

ii) unsaturated color is one with a mixture of white, and it contains various range of wavelength.

iii) RGB: grayscale is represented by R=G=B, e.g., gray can be represented to (100,100,100), the value of RGB should not be 0 or 255

CMY: grayscale is represented by C=M=Y, e.g., gray can be represented to (150,150,150) the value of CMY should not be 0 or 255

HSV: grayscale is represented along the vertical V axis, e.g., gray can be represented to (0,0,40) the value of V should not be 0 or 100, H=S=0

iv) we can decrease the saturation of a color by adding white into the color, which means decreasing the value that represents the color.

v) (0,50,50)

Question 2

a)

converted the true color into grayscale or index-based color.

Then set up a threshold value, which for instance, can be the middle value for grayscale like 127. If the index value for a specific color is higher than the threshold value, it will be classified as 1. Similarly, if the index value for a specific color is lower than the threshold value, it will be classified as 0.

b)

i) voiced sound. There are 2 reasons: voiced sound are almost periodic in spectrum, which means it's easier to analyze than unvoiced sound. Also, the black bar in spectrogram and the characteristic maxima in spectrum for voiced sound are formants, and different vowel have different formant location, so that formants can be powerful source of pronunciation information. Therefore, it's efficient to use formants in voiced sound to help transcript sound into text and map sound with language, which is the content of speech recognition.

ii) typical bit depths for speech: 8 bits. For music: 16 bits.

As for music, listeners use music to enjoy themselves, so they focus on the quality of music.

Therefore, the bigger bit depth for music, the higher quality for music, the more people like listening to it.
But for speech, listeners just want to understand the contents of speech, so the quality of speech sound is not so important compared to music. It doesn't matter for us to recognize the word through speech when using low bit depth. Therefore, 16 bits is good enough for people to get involved in the speech.
c)
i) utilize spatial property, i.e the image contains large areas of uniform color.
ii) The first row has 1600 pixel, so Log2(1600) =11 bit (2bytes) is used to represent the number of the pixel. True color (entirely red): 3byte. So, 5 bytes are used in total.
iii) True color (red): 3 bytes.
Because the first row has 1600 pixel and the bit depth for encoding the size is one byte (8 bit)
(red,255), (red,255), (red,255), (red,255), (red,255), (red,255), (red,70)
There will be 7 different runs, which means we need 7 bytes for the size
Plus 3 bytes for the pixel value
Therefore, (3+1) *7 = 28 bytes are used in total.

Question 3
a)
i) JPEG is lossy, because it throws away information during process, like the fine details in high frequency in DCT domain, and the color information through sub-sampled.
But because human eye is more sensitive to changes in light than in color, the color information thrown during sub-sampling won't be captured by human eyes. Also, those fine details in high frequency components are beyond the acuity of human perception. Therefore, some JPEG images look like the original image after decompression.
ii) the quantization factor is used to multiply on the quantization matrix, which means it scales the quantization values simultaneously and forms the quantization table. Then, we will use the whole quantization table to divide 8*8 blocks of DCT coefficients. Different quantization table (factor) are used according to the amount of compression needed.
For example, if we want to increase the compression rate, we can increase the quantization factor, but this will lead to loss of quality. If we want to decrease the compression rate, we can decrease the quantization factor.
iii) During this process we can separate luminance from chrominance, which helps us to apply different process to luminance and chrominance in next step of chroma sub-sampling. For example, luminance is going to be preserved completely while chrominance will be subsampled. The reason for that is the human eye is more sensitive to changes in luminance than in chrominance, so we can utilize this property to reduce color information and realize compression. Therefore, converting RGB to YCbCr can prepare for the following compression.
iv) DPCM will encode the difference between two values rather than encoding each value individually. Because DC coefficients are quite similar from one block to the next, so the difference will be very small. If we apply DPCM, the encoding will be economic.
v) store the quantization table.
Store the quantization factor.
Store the chroma-sub-sampling scheme.

If we use Huffman during the process, we still need to store the Huffman table.

b)

i)

Temporal redundancy means there is a high correlation between successive frames, and it can be used for inter-frame coding and performing motion estimation in MPEG--by utilizing the temporal redundancy, MPEG only encodes the changes from one frame to the next. For example, when a matching macroblock can be found in P and B frames, it helps to get higher temporal compression by just encoding the difference (error terms) and the motion vector.

Perceptual redundancy means there are some details beyond the acuity of human perception. In intra-frame coding, JPEG algorithm is used, which means treating the current key frame as a still image. And because it uses JPEG algorithm, it will execute steps of DCT, quantization and chroma sub-sampling. During those steps, it will discard high frequency part and some color information, which just utilize the property that human eyes can't see the abrupt change by high frequency component, and eyes are also less sensitive toward color than light. Therefore, if there is no matching macroblock can be found in frames, MPEG will use I frames and utilize the perceptual redundancy to do compression.

Spatial redundancy means elements that are duplicated within a structure, such as pixels in a still image, especially when there is only uniform color in a large area. Similar as perceptual redundancy, MPEG utilize it because of JPEG algorithm used in I frames production. In detail, it is because RLE steps in JPEG algorithm. Therefore, if there is no matching macroblock can be found in frames, MPEG will use I frames and utilize the spatial redundancy to do compression.

ii) Yes. The basic principle of fast motion estimation is that the number of search points is reduced by selectively checking only a small number of specific points. And for BMA, macroblocks are matched against blocks in previous or following frames within the square window of width (16 + 2w) pixels. That means the matching space is selectively reduced.

Question 4

a)

i) Frequency masking means a loud tone may mask a softer tone of similar or higher frequency. In this case, the threshold of hearing will be raised within a critical band in the presence of a masking tone.

ii) 20000/128 = 156. 156*128 = 19968. (20000-19968)/20000=0.0016

iii) 20000*0.3/128 = 46. 46/0.3*128 = 19626. (20000-19626)/20000=0.0187

iv) compressed rate: 20000*8 = 160000. 160000:128000=5:4

v) low frequency.

Human ability to distinguish between frequencies decreases nonlinearly from low to high frequencies. We can tell difference between pitches that are only a few Hz apart, while at high frequencies the pitches must be separated by more than 100 Hz before we notice a difference. Therefore, the range of frequency in which human cannot distinguish the difference are formed into critical bands. As we can see, bandwidth is larger when in higher critical band

and higher center frequency.

b)

unicast: transfer a message to only one recipient at one time. (point-to-point)

multicast: transmit to those users who desire to be transmitted.

Broadcast: transfer a message to all recipients simultaneously, no matter whether they are subscribers.

ii) source coding: source encoding refers to the compression of a digital signal. It means making the contents more compact as it can be compressed. e.g., Huffman coding; using MPEG-2

transport stream: a multiplex of elementary streams. It specifies a container format for the digital signal which can be used to encapsulate elementary streams.

Channel coding: a process of adding redundancy for detecting and correcting bit errors in digital communication systems. e.g., Reed-Solomon coding